

SMART QUEUE MANAGEMENT SYSTEM

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Abstract

The Smart Queue Management System is designed to efficiently monitor and manage crowd movement using ESP32 microcontroller and IR sensors. The system helps in reducing waiting time and avoiding overcrowding by continuously tracking the number of people in a queue in real time. The ESP32 enables wireless communication and IoT-based monitoring, allowing data to be accessed remotely through a web or mobile interface. IR sensors are used to detect entry and exit of individuals, ensuring accurate counting and smooth queue control. This system improves overall management in places such as banks, hospitals, ticket counters, and government offices by enhancing organization, reducing manual effort, and providing a smart, automated solution for crowd control..

Introduction

The Smart Queue Management System is designed to control and monitor queues in crowded places like banks, hospitals, and offices. It uses ESP32 and IR sensors to detect people entering and exiting the queue. The system helps in reducing waiting time, avoiding overcrowding, and improving overall efficiency through real-time monitoring and automation

Components Used

- ESP32



ESP32 is a Wi-Fi enabled microcontroller used for processing data and enabling IoT connectivity in the system.

- IR Sensors



IR sensors detect the presence of people by sensing obstacles using infrared signals.

- LEDs



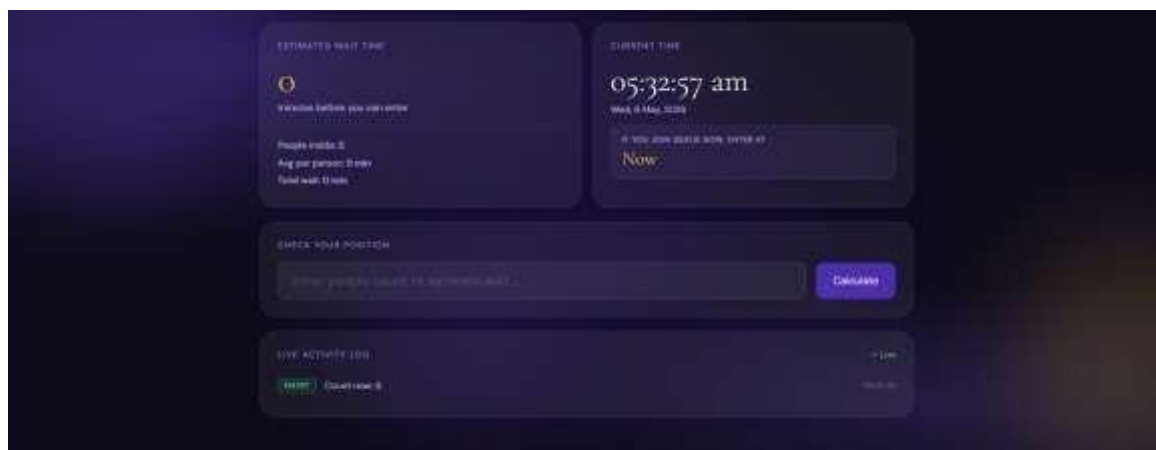
LEDs are used for indication of system status or queue conditions

- Firebase



Firebase is used to store and display real-time data online for remote monitoring.

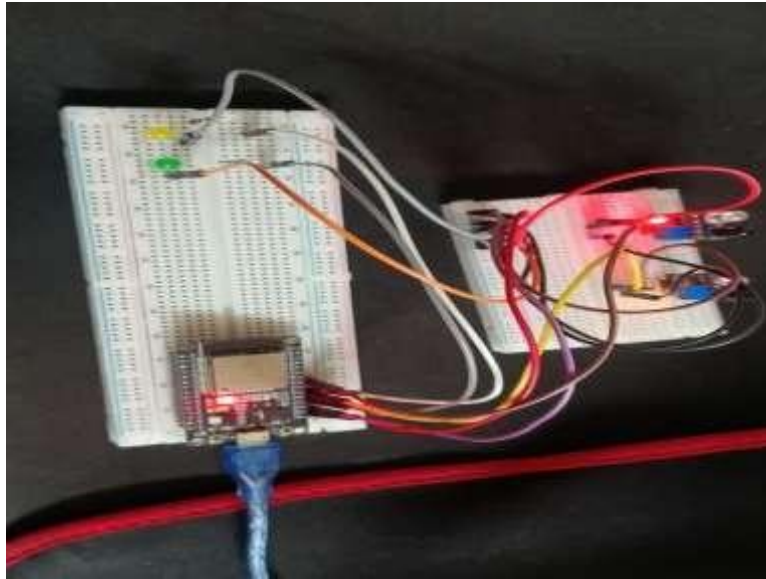
Website



Working Principle

The system uses IR sensors to detect people entering and exiting the queue. ESP32 processes the data and updates it in real time. This information is sent to Firebase for live monitoring, while LEDs indicate the queue status.

Project Setup



Advantages

- Real-time monitoring
- Low cost
- Easy to use

Conclusion

The Smart Queue Management System provides an efficient and automated way to monitor and control queues using ESP32 and IR sensors. It reduces crowding, minimizes waiting time, and improves overall management in public places. With Firebase integration, realtime data can be accessed remotely, making the system smart, reliable, and user-friendly.